Problem 1: Solve the following system for x, y, and z using Gauss-Jordan elimination.

$$\begin{aligned} \frac{1}{x} + \frac{2}{y} - \frac{4}{z} &= 1 \\ \frac{2}{x} + \frac{3}{y} + \frac{8}{z} &= 0 \\ -\frac{1}{x} + \frac{9}{y} + \frac{10}{z} &= 5 \end{aligned}$$

Solution

The augmented matrix in variables $\frac{1}{x}, \frac{1}{y}$ and $\frac{1}{z}$ is:

$$\left[\begin{array}{ccc|c} 1 & 2 & -4 & 1 \\ 2 & 3 & 8 & 0 \\ -1 & 9 & 10 & 5 \end{array}\right]$$

Make the first column below the pivot zero:

$$\left[\begin{array}{ccc|c} 1 & 2 & -4 & 1 \\ 0 & -1 & 16 & -2 \\ 0 & 11 & 6 & 6 \end{array}\right] \qquad \begin{aligned} R_2 &\to R_2 - 2R_1 \\ R_3 &\to R_3 + R_1 \end{aligned}$$

Make the second pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 2 & -4 & 1 \\ 0 & 1 & -16 & 2 \\ 0 & 11 & 6 & 6 \end{array}\right] \qquad R_2 \to -R_2$$

Make the second column above and below the pivot as zero:

$$\left[\begin{array}{ccc|c} 1 & 0 & 28 & -3 \\ 0 & 1 & -16 & 2 \\ 0 & 0 & 182 & -16 \end{array}\right] \qquad \begin{aligned} R_1 &\to R_1 - 2R_2 \\ R_3 &\to R_3 - 11R_2 \end{aligned}$$

Make the third pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 0 & 28 & -3 \\ 0 & 1 & -16 & 2 \\ 0 & 0 & 1 & \frac{-8}{91} \end{array}\right] \qquad R_3 \to \frac{1}{182} R_3$$

Make the third column above the pivot zero

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{-7}{13} \\ 0 & 1 & 0 & \frac{54}{91} \\ 0 & 0 & 1 & \frac{-8}{91} \end{array}\right] \qquad \begin{aligned} R_1 &\to R_1 - 28R_3 \\ R_2 &\to R_2 + 16R_3 \end{aligned}$$

Which gives us $\frac{1}{x} = -\frac{7}{13}$, $\frac{1}{y} = \frac{54}{91}$, $\frac{1}{z} = -\frac{8}{91}$. Hence the solution for variables x, y and z is:

Solution:

$$\boxed{x = -\frac{13}{7}, \quad y = \frac{91}{54}, \quad z = -\frac{91}{8}}$$

Problem 2: Solve the following system for x_1, x_2, and x_3 using Gauss-Jordan elimination.

$$\begin{aligned}
\frac{1}{x_1} + \frac{1}{x_2} + \frac{2}{x_3} &= 9 \\
\frac{2}{x_1} + \frac{4}{x_2} - \frac{3}{x_3} &= 1 \\
\frac{3}{x_1} + \frac{6}{x_2} - \frac{5}{x_3} &= 0
\end{aligned}$$

Solution

The augmented matrix in variables $\frac{1}{x_1}, \frac{1}{x_2}$ and $\frac{1}{x_3}$ is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right]$$

Make the first column below the pivot zero:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right] \qquad \begin{aligned} R_2 &\to R_2 - 2R_1 \\ R_3 &\to R_3 - 3R_1 \end{aligned}$$

Make the second pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & \frac{-7}{2} & \frac{-17}{2} \\ 0 & 3 & -11 & -27 \end{array}\right] \qquad R_2 \to \frac{1}{2}R_2$$

Make the second column above and below the pivot as zero:

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & \frac{-7}{2} & \frac{-17}{2} \\ 0 & 0 & \frac{-1}{2} & \frac{-3}{2} \end{array}\right] \qquad \begin{aligned} R_1 &\to R_1 - R_2 \\ R_3 &\to R_3 - 3R_2 \end{aligned}$$

Make the third pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & \frac{-7}{2} & \frac{-17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right] \qquad R_3 \to -2R_3$$

Make the third column above the pivot zero

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right] \qquad \begin{aligned} R_1 &\to R_1 - \frac{11}{2}R_3 \\ R_2 &\to R_2 + \frac{7}{2}R_3 \end{aligned}$$

Which gives us $\frac{1}{x1} = 1, \quad \frac{1}{x_2} = 2, \quad \frac{1}{x_3} = 3$. Hence the solution for variables x_1, x_2 and x_3 is:

Solution:

$$\boxed{x_1 = 1, \quad x_2 = \frac{1}{2}, \quad x_3 = \frac{1}{3}}$$

Problem 3: Solve the following system for x, y, and z using Gauss-Jordan elimination.

$$\begin{aligned} \frac{1}{x} + \frac{1}{y} + \frac{2}{z} &= 8 \\ -\frac{1}{x} - \frac{2}{y} + \frac{3}{z} &= 1 \\ \frac{3}{x} - \frac{7}{y} + \frac{4}{z} &= 10 \end{aligned}$$

Solution

The augmented matrix in variables $\frac{1}{x}, \frac{1}{y}$ and $\frac{1}{z}$ is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

Make the first column below the pivot zero:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right] \qquad \begin{aligned} R_2 &\to R_2 + R_1 \\ R_3 &\to R_3 - 3R_1 \end{aligned}$$

Make the second pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array}\right] \qquad R_2 \to -R_2$$

Make the second column above and below the pivot as zero:

$$\left[\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right] \qquad \begin{aligned} R_1 &\to R_1 - R_2 \\ R_3 &\to R_3 + 10R_2 \end{aligned}$$

Make the third pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right] \qquad R_3 \to \frac{1}{52}R_3$$

Make the third column above the pivot zero

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \qquad \begin{aligned} R_1 &\to R_1 - 7R_3 \\ R_2 &\to R_2 + 5R_3 \end{aligned}$$

Which gives us $\frac{1}{x} = 3, \quad \frac{1}{y} = 1, \quad \frac{1}{z} = 2$. Hence the solution for variables x, y and z is:

Solution:

$$\boxed{x = \frac{1}{3}, \quad y = 1, \quad z = \frac{1}{2}}$$

Problem 4: Solve the following system for x_1, x_2, and x_3 using Gauss-Jordan elimination.

$$\begin{aligned} \frac{2}{x_1} + \frac{2}{x_2} + \frac{2}{x_3} &= 0 \\ -\frac{2}{x_1} + \frac{5}{x_2} + \frac{2}{x_3} &= 1 \\ \frac{8}{x_1} + \frac{1}{x_2} - \frac{4}{x_3} &= -1 \end{aligned}$$

Solution

The augmented matrix in variables $\frac{1}{x_1}, \frac{1}{x_2}$ and $\frac{1}{x_3}$ is:

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & -4 & -1 \end{array}\right]$$

Make the first pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & -4 & -1 \end{array}\right] \qquad R_1 \to \frac{1}{2}R_1$$

Make the first column below the pivot zero:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -12 & -1 \end{array}\right] \qquad \begin{array}{l} R_2 \to R_2 + 2R_1 \\ R_3 \to R_3 - 8R_1 \end{array}$$

Make the second pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & -7 & -12 & -1 \end{array}\right] \qquad R_2 \to \frac{1}{7}R_2$$

Make the second column above and below the pivot as zero:

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{7} & \frac{-1}{7} \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & -8 & 0 \end{array}\right] \qquad \begin{array}{l} R_1 \to R_1 - R_2 \\ R_3 \to R_3 + 7R_2 \end{array}$$

Make the third pivot 1:

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{7} & \frac{-1}{7} \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 1 & 0 \end{array}\right] \qquad R_3 \to \frac{-1}{8}R_3$$

Make the third column above the pivot zero

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & \frac{-1}{7} \\ 0 & 1 & 0 & \frac{1}{7} \\ 0 & 0 & 1 & 0 \end{array}\right] \qquad \begin{array}{l} R_1 \to R_1 - \frac{3}{7}R_3 \\ R_2 \to R_2 - \frac{4}{7}R_3 \end{array}$$

Which gives us $\frac{1}{x1} = \frac{-1}{7}$, $\frac{1}{x_2} = \frac{1}{7}$, $\frac{1}{x_3} = 0$. Which implies that x_3 is not defined and hence the give system does not have solution in terms of x_1, x_2 and x_3.